

ULGI, Mongolia

WMO: 442140

Lat: 48.9800N

Lon: 89.9383E

Elev: 1715

StdP: 82.35

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.5	-27.8	-35.3	0.2	-28.1	-33.4	0.2	-25.4	13.9	-9.4	11.8	-9.0	0.5	270	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	27.1	13.4	25.5	12.7	24.0	12.1	15.6	23.5	14.3	22.3	13.3	21.0	4.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.5	11.2	18.1	11.1	10.2	17.4	10.0	9.4	17.0	50.0	23.3	45.9	22.2	43.0	20.9	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.4	9.2	DB	-32.0	29.7	3.3	1.7	-34.3	30.9	-36.3	31.9	-38.1	32.9	-40.5	34.1	(4)
(5)			WB	-31.7	16.4	3.2	2.5	-34.0	18.2	-35.9	19.6	-37.7	21.0	-40.1	22.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-17.0	-12.8	-5.1	3.7	9.9	16.3	18.0	15.9	9.3	1.6	-7.3	-14.2	(6)	
	DBStd	12.99	5.58	5.92	5.83	5.13	4.46	3.35	2.81	3.09	4.23	4.84	5.85	5.49	(7)	
	HDD10.0	3792	837	638	469	194	57	3	0	2	62	261	518	751	(8)	
	HDD18.3	6167	1096	872	727	439	262	76	41	87	272	518	768	1009	(9)	
	CDD10.0	720	0	0	0	5	54	191	247	183	39	1	0	0	(10)	
	CDD18.3	54	0	0	0	0	1	14	29	11	0	0	0	0	(11)	
	CDH23.3	475	0	0	0	0	8	118	232	110	7	0	0	0	(12)	
	CDH26.7	64	0	0	0	0	0	13	39	12	0	0	0	0	(13)	
(14) Wind	WSAvg	3.3	2.6	2.7	3.7	4.3	4.2	3.5	3.2	3.0	3.2	3.3	3.3	2.9	(14)	
(15) Precipitation	PrecAvg	107	1	1	1	5	10	27	32	22	9	3	1	2	(15)	
	PrecMax	164	5	3	10	15	42	170	148	85	42	13	12	12	(16)	
	PrecMin	54	0	0	0	0	0	3	4	0	0	0	0	0	(17)	
	PrecStd	27	1	1	2	4	8	25	22	18	8	3	2	2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.0	2.3	11.8	19.6	24.4	28.3	29.8	28.2	24.2	16.3	9.1	0.5	(19)	
		MCWB	-4.5	-2.7	3.3	7.1	11.2	12.9	15.0	14.2	11.4	6.8	2.8	-3.6	(20)	
	2%	DB	-4.2	0.0	9.1	16.3	21.8	26.2	27.6	26.0	21.3	13.4	6.1	-2.2	(21)	
		MCWB	-7.2	-4.3	1.8	5.7	10.0	12.3	13.7	13.0	10.0	5.4	0.7	-5.5	(22)	
	5%	DB	-6.4	-1.9	6.9	14.0	19.7	24.5	25.8	24.3	19.2	11.5	3.7	-4.0	(23)	
		MCWB	-8.9	-5.6	0.2	4.5	8.9	11.7	13.2	12.4	9.1	4.3	-0.9	-6.9	(24)	
	10%	DB	-8.6	-3.9	4.6	12.0	17.8	22.9	24.2	22.6	17.2	9.6	1.3	-6.0	(25)	
		MCWB	-10.8	-7.2	-1.1	3.5	7.8	11.1	12.6	11.7	8.1	3.2	-2.7	-8.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.5	-2.3	4.4	8.2	12.6	15.4	17.7	16.6	12.9	7.8	3.6	-3.4	(27)	
		MCDB	-1.2	2.0	10.4	17.2	22.4	24.8	25.8	25.1	21.3	13.8	8.9	0.2	(28)	
	2%	WB	-6.9	-4.1	2.1	6.3	11.0	13.8	16.1	15.1	11.1	6.0	1.3	-5.2	(29)	
		MCDB	-4.2	-0.2	8.3	14.7	19.9	22.5	23.0	23.2	18.9	12.7	6.0	-2.1	(30)	
	5%	WB	-8.7	-5.5	0.4	5.1	9.7	13.0	14.8	13.7	9.8	4.7	-0.7	-6.6	(31)	
		MCDB	-6.3	-1.9	6.6	12.9	17.8	20.8	22.9	21.4	17.7	10.7	3.6	-3.8	(32)	
	10%	WB	-10.6	-7.1	-1.0	4.0	8.4	12.2	13.8	12.7	8.6	3.3	-2.4	-8.2	(33)	
		MCDB	-8.4	-3.8	4.4	11.1	16.8	20.3	21.3	19.9	16.4	9.3	1.5	-5.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	12.2	12.7	11.9	11.0	10.5	10.1	10.9	11.9	11.2	9.5	8.6	(35)	
		MCDBR	10.4	12.6	14.1	13.9	13.6	12.7	12.4	13.5	14.7	14.0	11.2	9.3	(36)	
	5% WB	MCWBR	8.4	9.6	9.1	7.6	6.9	4.7	4.6	5.0	7.1	8.4	7.6	7.2	(37)	
		MCWBR	10.5	12.3	13.4	12.6	12.0	11.0	10.8	11.4	13.2	13.0	11.0	9.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.233	0.260	0.306	0.352	0.346	0.350	0.347	0.328	0.292	0.277	0.254	0.235	(40)	
		taud	2.367	2.310	2.230	2.203	2.278	2.351	2.383	2.409	2.469	2.444	2.382	2.338	(41)	
	Ebn at Noon		858	903	904	893	912	908	905	910	918	874	817	799	(42)	
		Edh at Noon	64	87	114	131	127	119	114	105	89	76	63	58	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.66	2.75	4.22	5.47	6.35	6.53	6.33	5.67	4.54	2.81	1.68	1.28	(44)	
	RadStd		0.07	0.10	0.20	0.29	0.38	0.33	0.36	0.23	0.27	0.22	0.12	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-1.43	-1.63	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	-2.00	N/A	-0.73	-1.06	N/A	N/A	N/A	N/A

Nomenclature: See separate page